Mixed Mental Maths

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Easy / Standard

1. 54

2. B — 4/10 = 0.4

3. 2,350

4. 42

5. C — 48 is divisible by 2

6. 18

7. 120 minutes

8. £2.60

9. A, C and E (16, 20 and 24 are all multiples of 4 (4×4, 4×5, 4×6); 18 and 22 are not)

10. 62

11. 15 (20÷4=5, 5×3=15)

12. B — A square has 4 equal sides and 4 right angles

13. 63

14. -4°C (2 − 6 = −4)

15. 650

Standard

16. 60

17. 0.6

18. A — 45/100 = 9/20

19. 6,800

20. 12

21. 2,500 g

22. C — 29 has no factors other than 1 and 29

23. 12 (18÷3=6, 6×2=12)

24. 26 cm (2×(8+5)=26)

25. 11:10

26. A, B and D (2/4, 3/6 and 5/10 all simplify to 1/2; 2/5 and 4/9 do not)

27. 600

28. 3 (it appears most often)

29. B — 0.45 < 0.5 < 0.6

30. 30 km/h (60÷2=30)

Standard / Hard

31. 28 (1/2 of 50 = 25; 25 + 3 = 28)

32. C — 10% of £20 = £2; £20 − £2 = £18

33. 21 (30 − 9 = 21)

34. XXIV (XX + IV)

35. 30 cm (1.2 m ÷ 4 = 0.3 m = 30 cm)

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36. C — 48 ÷ 6 = 8

37. 300 g (200÷4=50 g per person; 50×6=300 g)

38. 3°C (−4+9=5; 5−2=3)

39. A, B, D and E (1/2=50%, 1/5=0.2, 0.25=25% and 3/4=0.75 are all correct; 2/5 (0.4) is actually less than 1/2 (0.5), not greater)

40. C — 7×5=35

41. 10:55

42. C — Each term is multiplied by 3: 54×3=162

43. £3.65 (£10.00 − £6.35 = £3.65)

44. 24 (40÷5=8, 8×3=24)

45. 5 km/h (20 km ÷ 4 hours = 5 km/h)

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Homework Worksheet — Answers

Easy / Standard

1. 56

2. B — 3/10=0.3

3. 4,170

4. 62

5. C — 51 is not divisible by 2

6. 25

7. 180 minutes

8. £2.25

9. A, C and E (15, 25 and 40 are all multiples of 5; 18 and 32 are not)

10. 53

11. 12 (16÷4=4, 4×3=12)

12. B — A triangle has 3 sides

13. 54

14. -4°C (3 − 7 = −4)

15. 380

Standard

16. 66

17. 0.7

18. A — 20/100 = 1/5

19. 5,400

20. 9

21. 1,500 ml

22. C — 23 has no factors other than 1 and 23

23. 10 (25÷5=5, 5×2=10)

24. 28 cm (4×7=28)

25. 10:15

26. A, B and D (2/8, 3/12 and 5/20 all simplify to 1/4; 2/6 and 3/9 both simplify to 1/3)

27. 750

28. 6 (in order: 2, 4, 6, 9, 11 — the middle number is 6)

29. B — 0.3 < 0.35 < 0.4

30. 30 km/h (90÷3=30)

Standard / Advanced

31. 24 (half of 30 = 15; 15 + 9 = 24)

32. 20 (40 ÷ 4 = 10; 10 × 2 = 20)

33. 22 (4 × 8 = 32; 32 − 10 = 22)

34. 4 (first three even numbers: 2, 4, 6; mean = 12 ÷ 3 = 4)

35. 1 (5 × 9 = 45; 45 ÷ 4 = 11 remainder 1)

36. 32 m² (inner dimensions 8 m × 4 m)

37. C) £20 (cost per toy = £60÷20 = £3; profit per toy = £4−£3 = £1; total profit = 20×£1 = £20)

38. 80 (total needed for average 80 over 4 tests = 320; so far 70+80+90=240; 320−240=80)

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39. C) 16 minutes (first 6 minutes drains 30 litres, leaving 30 litres; remaining time = 30÷3 = 10 minutes; total = 6+10 = 16)

40. 12 hectares (scale 1:10,000 means 1 cm = 100 m; actual dimensions 300 m × 400 m = 120,000 m^2 = 12 hectares)

41. C) 30 (ratio parts 1:2:3 with largest 10 more than smallest: 3x = x+10, so x=5; numbers 5, 10, 15; sum=30)

42. A, B and D are true (C is false since the average cost per pupil, £215÷20=£10.75, is not more than £11; E is false since 10% of £200 is £20, which is more than the £15 coach fee)

43. 10:40 am (riding time 30÷20=1.5 hours = 90 minutes; plus a 10-minute break = 100 minutes after 09:00)

44. C) 3 (original number = 36 ÷ 1.2 = 30; 10% of 30 = 3)

45. 54 cm^2 (side length = 3 cm since $3^3 = 27$; surface area = $6 \times 3^2 = 54$ cm^2)

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